MSMS 408 : Practical 05

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Question

Use rejection sampling technique to generate random sample from a distribution with PDF

$$f(x) = 20 \cdot x(1-x)^3 \cdot I_{(0,1)}(x)$$

Algorithm

Suppose we have to generate random sample from $p(x)$, we call it our **target distribution**. We choose a **proposal distribution** $q(x)$, say.

I. Generate $x_i \sim q(x)\ \forall\ i = 1(1)n$.

II. Generate $u_i \sim U(0,1)\ \forall\ i = 1(1)n$.

III. Accept x_i if

$$u_i \leq \frac{p(x_i)}{M \cdot q(x_i)}$$

where $M > 0$ is a suitable constant so that $p(x) \leq M \cdot q(x)$ is satisfied.

Accepted samples follow the target distribution $p(x)$.

Here, we are given that $p(x) = 20 \cdot x(1-x)^3$. We take $q(x) = 1 \cdot I_{(0,1)}(x)$ *i.e.* $U(0,1)$ distribution.

To find constant M recall that we need $p(x) \leq M \cdot q(x) \Rightarrow p(x) \leq M$. Notice, $p(x)$ is the PDF of a $\beta_1(2,4)$ distribution which has mode

$$\frac{2-1}{2+4-2} = \frac{1}{4}.$$

So $p(0.25) \approx 2.11$ is a suitable choice for M.

R Program

```r
set.seed(14)
```

```r
n <- 2000
```

```r
p <- function(x) 20 * x * (1-x)^3
```

```r
q <- function(x) 1
```

```r
M <- 2.11
```

```r
q1 <- function(x) M * q(x)
```

```r
x <- runif(n)
y <- runif(n, min = 0, max = q1(x))
```

```r
good <- y <= p(x)
accepted <- x[good]
```

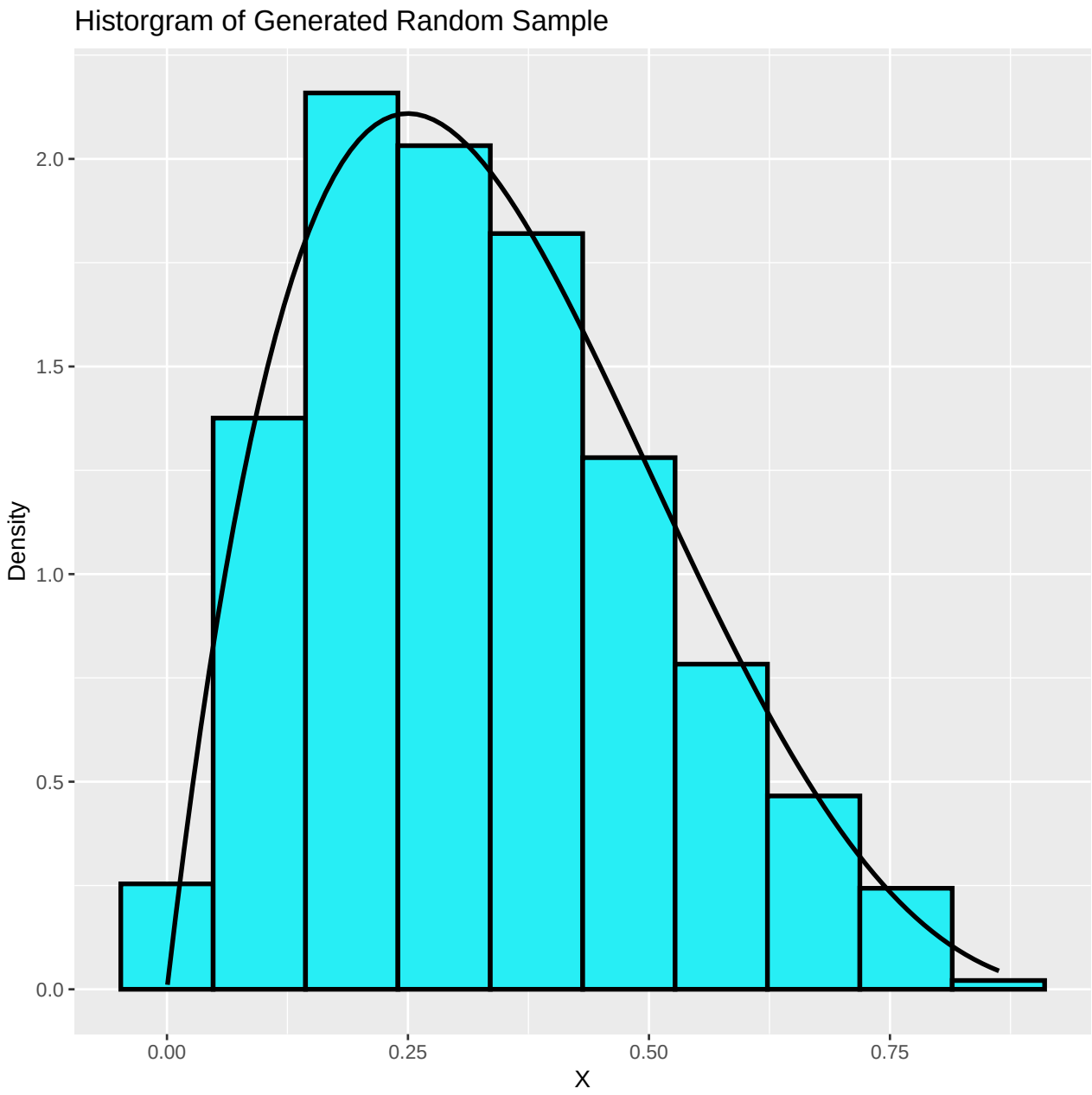
```r
count_accepted <- length(accepted) # or sum(good)
count_accepted

## [1] 986
```

Thus we have a sample of size 986 from $p(x)$.

The **acceptance rate** was 0.493 whereas the **theoretical acceptance rate** is 0.4739336.

Let us have a histogram of the generated sample.

```
data.frame(accepted) %>%
  ggplot(aes(x = accepted)) +
  geom_histogram(aes(y = after_stat(density)), bins = 10,
                 linewidth = 1, color = "black", fill = "#27EEF5") +
  stat_function(fun = p, linewidth = 1) +
  labs(x = "X", y = "Density",
       title = "Historgram of Generated Random Sample")
```

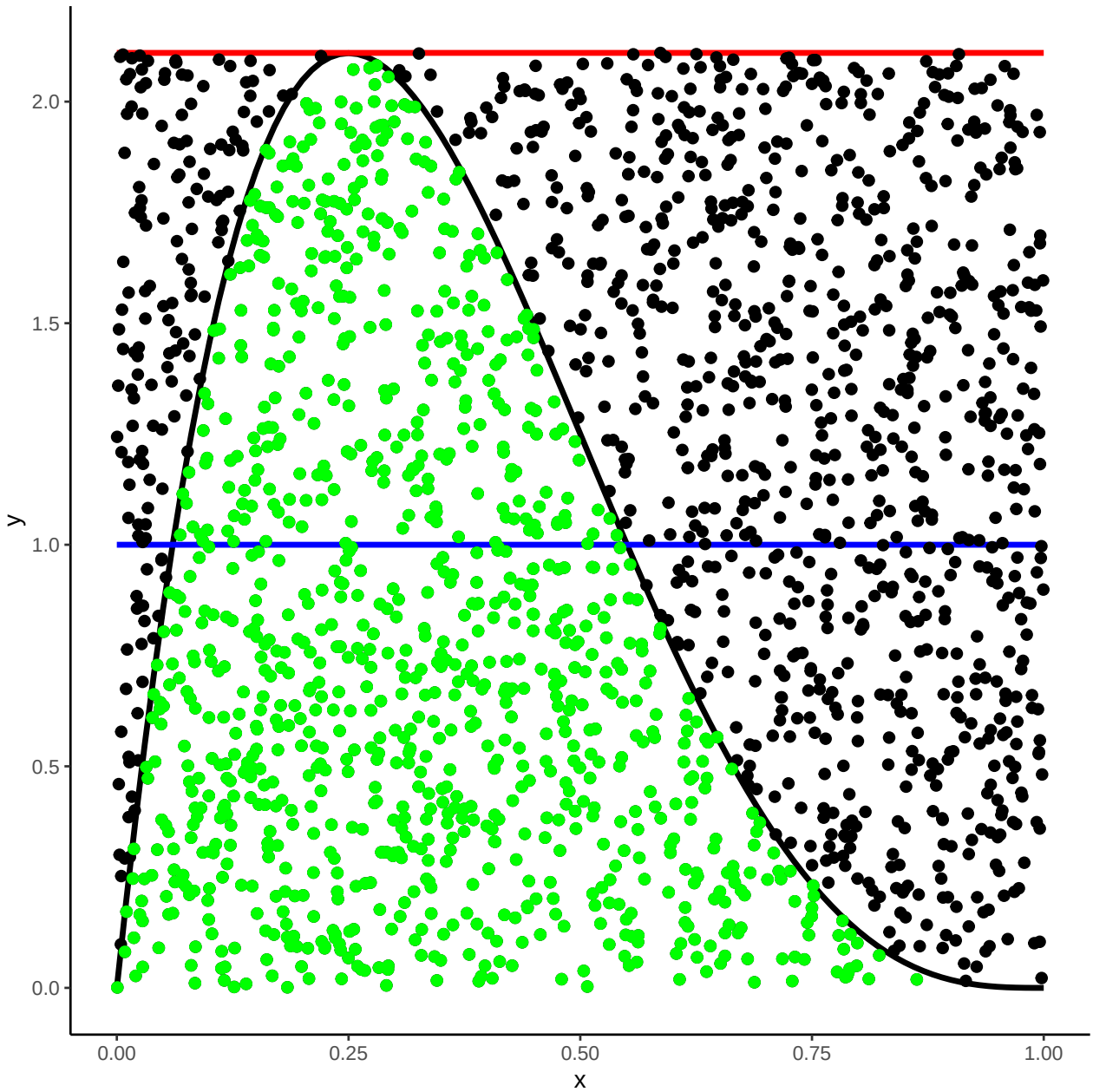


➔ Conclusion

✎ The actual density curve fits the histogram well.

Let us have a simple visualization of the rejection sampling technique.

```
data.frame(x = c(0, 1)) %>%
  ggplot() +
  stat_function(aes(x = x), fun = p, linewidth = 1.25) +
  theme_classic() +
  stat_function(aes(x = x), fun = q, linewidth = 1.25, col = "blue") +
  stat_function(aes(x = x), fun = q1, linewidth = 1.25, col = "red") +
  geom_point(data = data.frame(x = x, y = y), aes(x = x, y = y), size = 2) +
  geom_point(data = data.frame(x = x[good], y = y[good]),
             aes(x = x, y = y), size = 2, col = "green")
```



✎ The accepted points are shown in green, these points lie under the envelop as well as the target density.